HOTEL BOOKING DATA ANALYSIS

Final Project Report

Python • Pandas • Matplotlib • Jupyter Notebook

Dataset Overview	Result
Total records	10,000
Usable columns	18
Missing values after cleanup	0
Duplicate rows	0
Date range in dataset	2022–2023

This report presents cleaned summaries and consistently sized charts. Chart values and labels have been formatted to improve readability.

1. Project Overview

The goal of this project is to explore hotel booking data and summarize booking status, room demand, payment preferences, customer characteristics, discounts and revenue.

2. Objectives

• Clean the dataset and check missing values and duplicate rows.
• Summarize booking status, room type, payment mode and gender.
• Review customer age, discounts and revenue patterns.
• Present findings with clear charts and actionable recommendations.

3. Tools Used

Python, Pandas, Matplotlib and Jupyter Notebook.

4. Data Cleaning

Removed fully empty columns. The cleaned dataset contains 10,000 rows and 18 columns, with 0 missing cells and 0 duplicate rows detected.

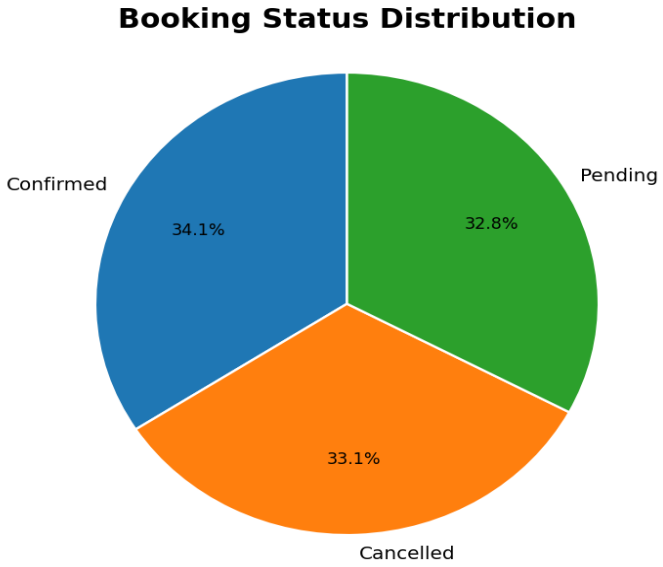
5. Key Findings

• Booking status — Confirmed: 3,414; Cancelled: 3,309; Pending: 3,277.
• Most common room type — Super Luxury (1,507 records).
• Most common payment mode — Credit Card (2,054 records).
• Customer types — Existing: 5,086; New: 4,914.
• Average customer age — 48.92 years.
• Average discount — 25.30; range: 0–50.
• Average Total Revenue per record — 25,048,881.14. Currency/unit is not specified in the dataset.
• Highest city revenue — Nur-Sultan (23,985,555,294).

6. Charts and Interpretation

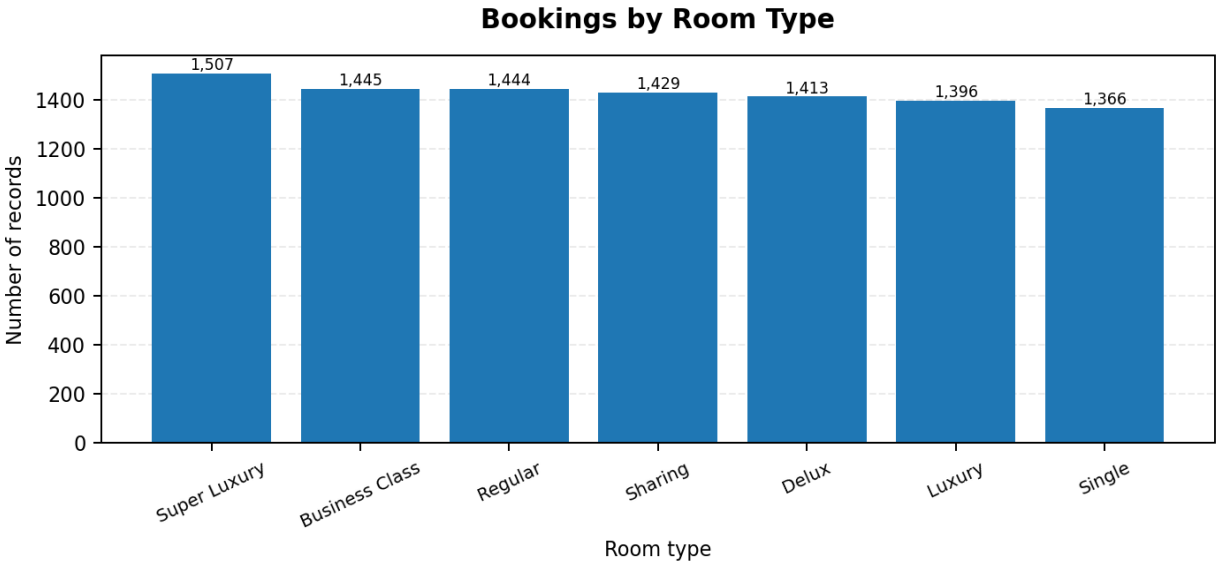
Charts are kept at a consistent width and height. Large revenue values are abbreviated on the axes (M = million, B = billion); exact values are written below each chart where relevant.

Booking Status Distribution



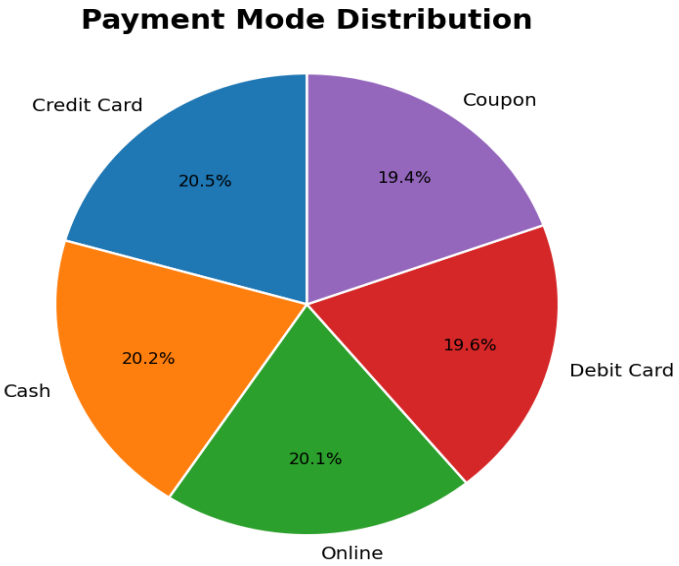
Confirmed: 3,414 (34.14%), Cancelled: 3,309 (33.09%), Pending: 3,277 (32.77%).

Bookings by Room Type



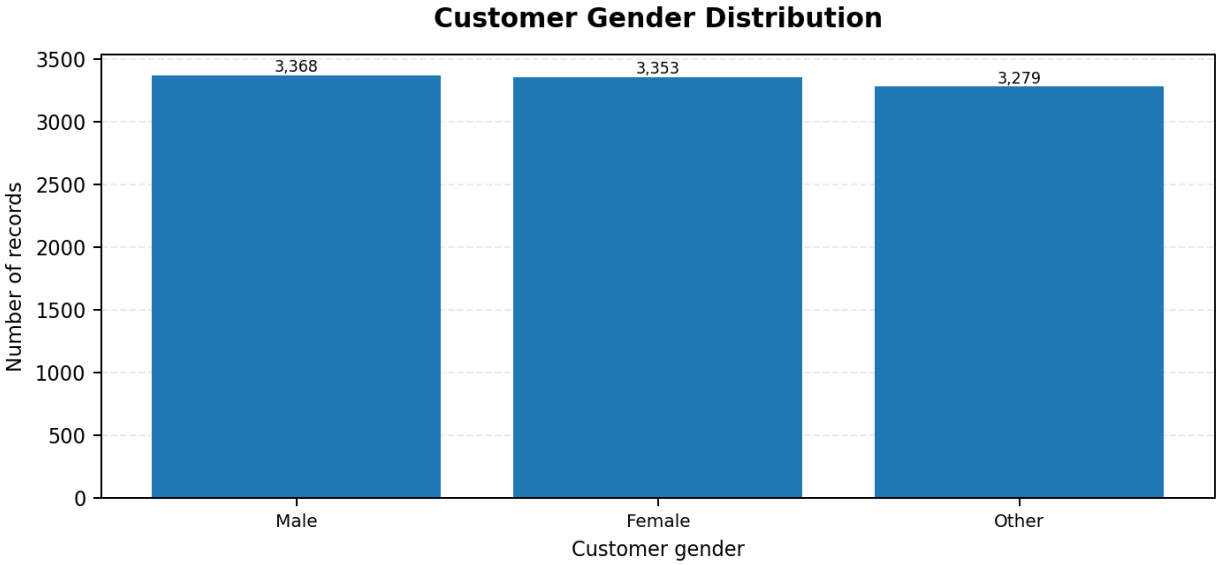
Most frequent room type: Super Luxury (1,507 records).

Payment Mode Distribution



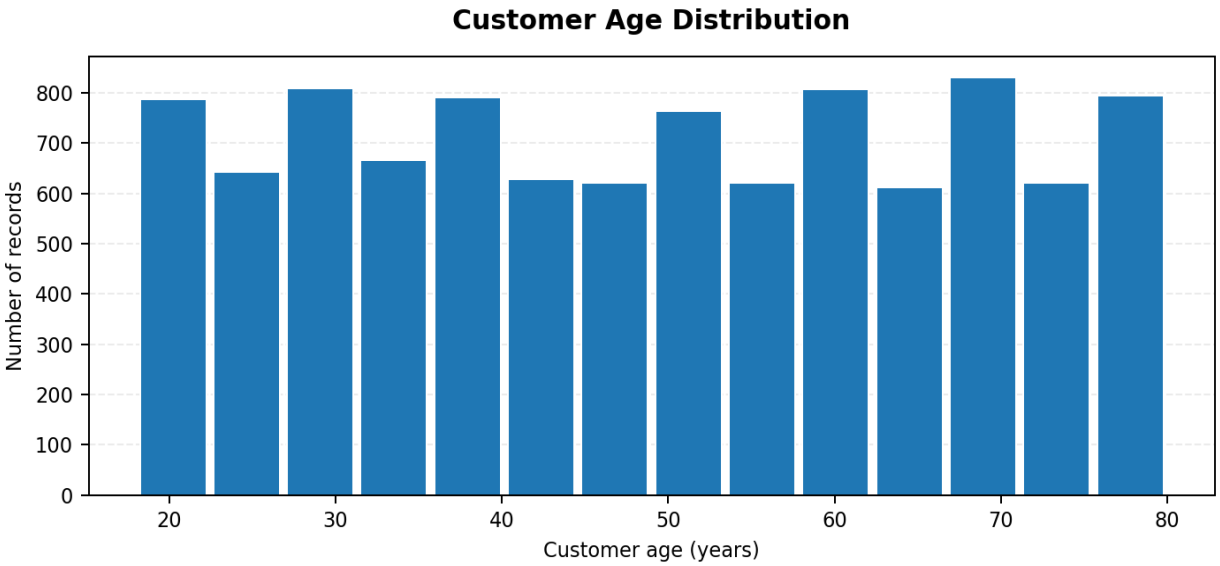
Most common payment mode: Credit Card (2,054 records).

Customer Gender Distribution



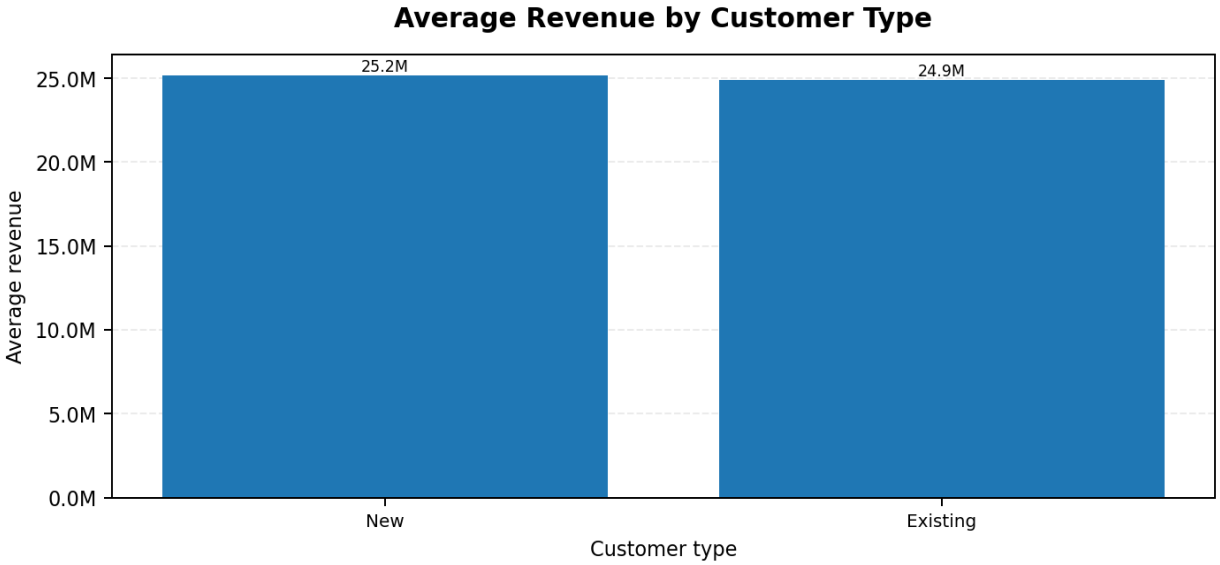
Male: 3,368; Female: 3,353; Other: 3,279.

Customer Age Distribution



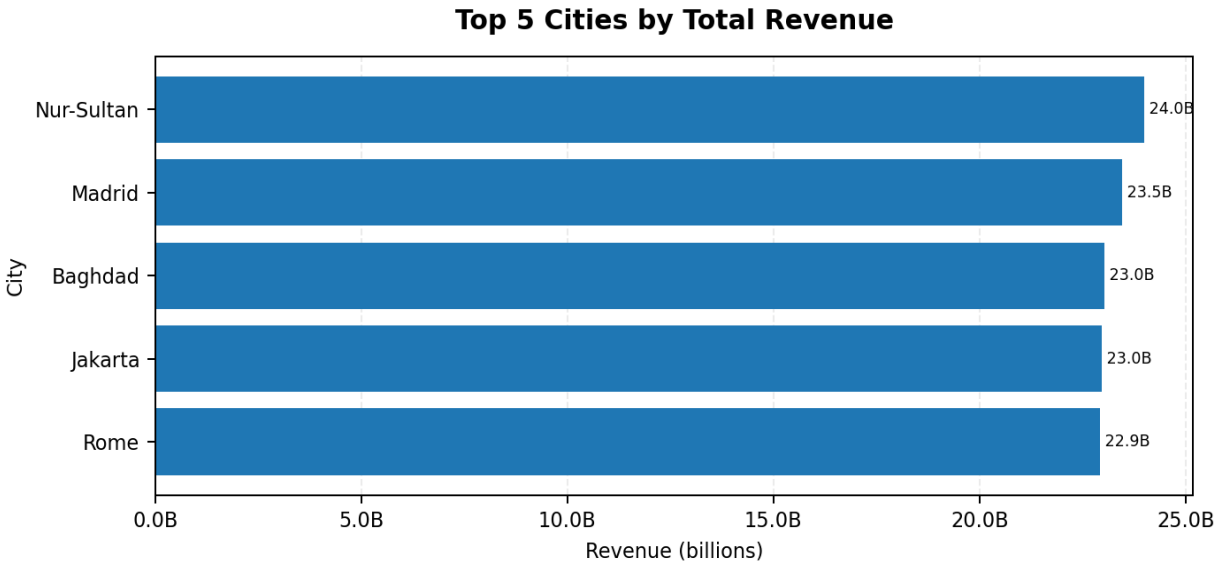
Average age: 48.92 years; age range: 18–80 years.

Average Revenue by Customer Type



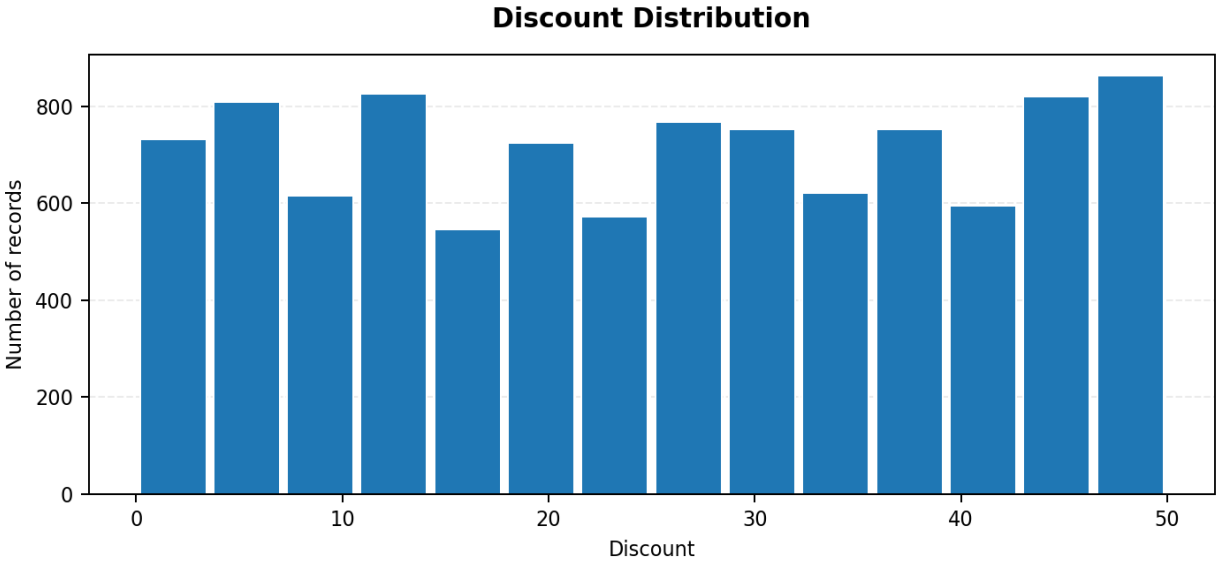
New: 25,179,006.05; Existing: 24,923,156.84.

Top 5 Cities by Total Revenue



Highest total revenue among the top five: Nur-Sultan (23,985,555,294).

Discount Distribution



Average discount: 25.30; range: 0–50.

7. Recommendations

• Investigate cancellation reasons by city, room type, customer type and payment mode.
• Review pending bookings and follow up where appropriate.
• Compare revenue with booking counts and room availability before making pricing decisions.
• Evaluate whether discounts improve confirmed bookings and revenue.
• Confirm the currency and definition of Total Revenue before financial interpretation.

8. Limitations

This is a descriptive analysis based only on the columns supplied. Revenue units are not defined, and observed relationships do not establish cause and effect. The dataset covers 2022–2023, so it may not reflect current hotel performance.

9. Conclusion

The project demonstrates how Python can be used to clean, summarize and visualize hotel booking records. The charts highlight booking-status balance, room-type demand, payment preferences and revenue differences, providing a basis for further investigation.